A Privacy-Preserving Federated Learning Approach for Primary Colorectal Malignancy Detection, Tumor Multiplicity Estimation and Regional Occurrence Mapping

Abstract:
Colorectal cancer (CRC) is still one of the most common and deadly types of cancer in the world, and patient outcomes are greatly enhanced by early detection. However, collaborative development of AI-driven diagnostic tools across institutions is often hindered by privacy concerns and regulatory restrictions on data sharing. We propose a **privacy-protecting federated learning (FL) framework** designed to detect the **first primary malignant colorectal tumor**, estimate the **tumor multiplicity**, and map the **regional occurrence of tumors** in this study. Leveraging the SEER (Surveillance, Epidemiology, and End Results) dataset, we structured our solution around three task-specific pipelines using deep learning approaches. Each model was trained locally on different clinical feature subsets, and global learning was achieved through federated averaging without exposing raw patient data. The deep neural network achieved strong performances having accuracies **96.1%**, **92.7%** & **97.5%** respectively in malignancy detection, tumor count prediction and regional tumor mapping. Our federated system enables secure and effective collaboration between medical institutions, enhancing early cancer detection and treatment planning while ensuring patient data confidentiality. This study demonstrates that federated learning is feasible for large-scale, privacy-sensitive cancer diagnostics and opens avenues for real-world deployment in clinical environments.
Keywords*—Federated Learning, Colorectal Cancer, SEER Dataset, Privacy-Preserving AI, Tumor Multiplicity Estimation, Regional Tumor Mapping, Medical Machine Learning, Detection.*

I. Introduction:

Globally, colorectal cancer is one of the main causes of cancer-related mortality. Catching it early makes a huge difference in how well patients can respond to treatment. That's why detecting the first occurrence of malignant colorectal cancer, understanding how many tumors a patient has, and knowing exactly where these tumors are located is so important for doctors to ensure effective care. Today, hospitals and research centers are collecting massive amounts of health data. Using this data to build strong cancer detection models is promising, but sharing sensitive patient information across institutions raises serious privacy concerns. Directly pooling patient data isn't always possible or ethical. This is where federated learning comes in—a technique that enables the cooperative training of AI models by several universities without the need to share raw data. The data stays safely on-site, while the models learn together, preserving patients' privacy. In our study, we propose a privacy-preserving federated learning approach that can detect the first primary colorectal malignancy, estimate the total tumor burden, and map the regional occurrence of tumors. By leveraging clinical data from different hospitals in a secure way, our method aims to improve early cancer detection and support better treatment decisions—all while keeping patient information confidential.

II. Background Study:

Cancer diagnosis and prognosis have traditionally relied on centralized data analysis, where patient records, imaging data, and clinical outcomes are aggregated at a single site for machine learning model training. While this approach can yield highly accurate models, it raises significant ethical and legal concerns regarding data privacy, particularly in healthcare environments governed by regulations such as HIPAA and GDPR To address these challenges, **federated learning (FL)** has emerged as a promising decentralized paradigm that enables collaborative model training across multiple institutions without sharing raw patient data [1], [2].

Recent research demonstrates the growing momentum of FL in healthcare applications. Munawar Hossain et al. [1] implemented a federated deep learning framework for lung and colon cancer detection using the LC25000 histopathology dataset, leveraging Inception-V3 as the backbone. Their model achieved near-perfect accuracy while maintaining privacy but faced real-world scalability and computational limitations. Similarly, Rodrigues et al. [12] validated the robustness of FL on heterogeneous histopathological data, showing FL's superior generalization compared to centralized learning, particularly in cross-domain settings.

In the broader healthcare context, Syed Raza Abbas et al. [2] provided an extensive review on FL applications in smart healthcare, highlighting its synergy with IoT for predictive diagnostics and remote monitoring. Their findings underline the importance of privacy-preserving training but point to ongoing challenges such as communication overhead, data heterogeneity & susceptibility to adversarial attacks—challenges particularly relevant to federated medical AI.

Several domain-specific FL implementations further underscore the potential of FL in cancer diagnostics. Maithili Jha et al. [5] proposed **PrivFED**, which incorporated SMOTE, PCA, and CatBoost into an FL pipeline for breast cancer detection, achieving 99.95% accuracy at the edge. While effective, the approach raised concerns about overfitting due to synthetic data and the need for stronger privacy enforcement. Likewise, Ananya Ghosh and Parthiban Krishnamoorthy [3] introduced **FedADC**, a federated method integrating knowledge distillation for waste classification, which, although not directly medical, offers architectural insights relevant for privacy-constrained data environments like cancer registries.

Yuxin Miao et al. [4] presented **FedSAF**, a federated method combining attentive message passing and Fisher Information Matrix-based strategies for gastric cancer detection. Achieving up to 98.43% accuracy, the study emphasized the benefits of personalization in federated systems. Meanwhile, Subramanian et al. [13] explored optimization in FL by comparing **FedAvg** and **FedProx** in CT and MRI-based cancer detection, showing improved convergence with FedProx under non-IID data conditions—a frequent challenge in multi-institutional medical data sharing.

Beyond individual implementations, systematic reviews by Ankolekar et al. [6], [9] and Reddy et al. [11] consolidate findings across dozens of studies on FL in oncology. These reviews consistently affirm FL's ability to enhance model generalization and privacy preservation while pointing out the lack of standardization, reproducibility issues, and the absence of detailed reporting on privacy techniques.

In addition to image-based datasets, recent works have explored FL with structured and registry data. Tang and Wan [16], for instance, developed a prognostic model using the SEER database for papillary renal cell carcinoma, demonstrating how clinical features like TNM staging and surgical history can be effectively used in predictive modeling. Lin-Feng et al. [17] similarly used SEER data to study conjunctival tumors, extracting population-level trends and identifying survival factors. These studies validate the SEER dataset's utility in population-based cancer research and highlight

its relevance for FL applications using structured clinical data rather than images.

Furthermore, FL implementations with built-in security enhancements are increasingly emerging. Sharia Arfin Tanim et al. [7] implemented homomorphic encryption and attention-based fusion in their Parkinson's disease FL model, ensuring both strong data protection and effective learning. Likewise, Shubhi Shukla et al. [8] integrated **Differential Privacy** into their breast cancer Federated Learning pipeline, achieving a privacy-preserving learning setup with a strong privacy budget, albeit with some trade-offs in accuracy.

Comparative FL studies, such as the one conducted by Osei et al. [20], evaluated multiple FL strategies (e.g., Per-FedAvg, FedOpt) on lung cancer data, demonstrating that while FL can nearly match centralized performance, personalization and data imbalance remain hurdles. Lastly, Mahdi et al. [15] explored explainable AI with deep learning models for early colon cancer detection, emphasizing the importance of interpretability in clinical applications—an aspect that could complement federated systems where black-box models raise concerns.

Collectively, the literature underscores that while federated learning holds strong promise for privacy-preserving, multi-institutional cancer diagnosis, practical challenges remain. These include data imbalance, computational costs, adversarial vulnerability, and model personalization across heterogeneous clinical environments. Nonetheless, with its potential to improve early detection and treatment planning—particularly for cancers like colorectal cancer that benefit from early diagnosis—FL represents a pivotal shift toward secure, collaborative medical AI

III. Literature Review:

Serial No.	Author & Year	Dataset / Domain	Model / Techniques	Accuracy / AUC
1	Md. Munawar Hossain et al., 2024	LC25000 (Lung & Colon Cancer)	Inception-V3, Federated Learning	99.87% (lung), 100% (colon)
2	Syed Raza Abbas et al., 2024	Review (Smart Healthcare + IoT)	Federated Learning (FL), Privacy/Security Discussion	Not specified
3	Ananya Ghosh & P. Krishnamoorthy, 2024	Waste Image Dataset (Upsampled)	FedADC (FedAvg + KD + MCL), DenseNet201	85.56%
4	Yuxin Miao et al., 2025	SEED, BOT, CIFAR-10, FashionMNIST	FedSAF (AMP + FIM + Model Splitting)	Up to 98.43%
5	Maithili Jha et al., 2024	Wisconsin Breast Cancer Dataset	PrivFED (SMOTE + Isolation Forest + PCA + CatBoost)	99.95% (edge), 98% (central)
6	Anshu Ankolekar et al., 2024	Systematic Review (25 Oncology Studies)	Federated Learning in Oncology	Up to 99.69%
7	Sharia Arfin Tanim et al., 2024	PD-BioStampRC21 (Parkinson's Disease)	Attention-based Fusion Model + CKKS Encryption	Up to 91.5%
8	Shubhi Shukla et al., 2025	Breast Cancer Wisconsin Dataset	FL + Differential Privacy ($\varepsilon = 1.9$)	96.1%
9	Ankolekar et al., 2025	Review (Breast, Lung, Prostate Cancer)	Blockchain-integrated FL	95.95%–99.69%
10	Chai et al., 2024	Four Cancer Datasets	AdFed (Decentralized FL + Gene Feature Analysis)	0.605 (Acc), 0.588 (AUC)
11	Reddy et al., 2025	Review (Colorectal & Breast Cancer)	Multiple FL Studies (Systematic Review)	88.5%, 0.92 AUC
12	Rodrigues et al., 2025	LC25000 (Lung & Colon Cancer)	SqueezeNet (Central vs FL)	99.32%
13	Subramanian et al., 2022	CT/MRI (Cervical, Lung, Colon Cancer)	FedAvg vs FedProx (Bayesian Opt.)	Up to 83.31%
14	Barbosa et al., 2025	BreakHis (Breast Histopathology)	AlexNet, ResNet50 (FL vs Centralized)	Up to 97.91%
15	Mahdi et al., 2024	LC25000 (Colon Cancer)	VGG-16, ResNet50, Custom CNN + LIME, DeepLIFT	Up to 96.7%
16	Tang and Wan, 2025	SEER (pRCC)	Nomogram Model (Age, TNM, Grade, etc.)	AUC: 0.798, 0.781, 0.754
17	Lin-Feng et al., 2025	SEER (1975–2018, PMCT cases)	Multivariate Cox Regression	Not applicable
18	Ciobotaru et al., 2025	Systematic Review (DL + FL in Breast Cancer)	ResNet, ViT, U-Net	Not directly reported
19	Gupta et al., 2025	Theoretical (AI in Healthcare)	FL in diagnostics, imaging, drug discovery	Not reported
20	Osei et al., 2025	LC25000 (Lung Cancer)	Per-FedAvg, FedOpt, FedBn; Grad-CAM	99.07%, AUC 0.9986

IV. Methodology:

This study presents a privacy-protecting federated learning framework for multi-faceted colorectal cancer analysis using the SEER (Surveillance, Epidemiology, and End Results) dataset. The proposed system addresses three core tasks: (1) detection of the first primary malignant colorectal tumor, (2) estimation of tumor multiplicity, and (3) mapping the regional tumor occurrence. The methodology includes data preparation, feature engineering, model selection, and federated model training. Each task is implemented using a dedicated model pipeline, and the system is designed to support collaborative training across multiple institutions without data sharing.

A. Data Source and Feature Segmentation

The clinical data used in this study were obtained from the SEER program, comprising thousands of anonymized colorectal cancer patient records. This dataset has a wide variety of features, including data (e.g., age, sex, race), tumor histology, staging information, tumor count, and location-specific codes such as "TNM 7/CS v0204+ Schema recode".
To support the multi-task framework, we segmented the dataset into three task-specific feature subsets:

Feature Set 1: Relevant to identifying whether a tumor is the first primary malignancy.
Feature Set 2: Relevant to predicting the number of in situ or malignant tumors a patient has.
Feature Set 3: Focused on determining the tumor's anatomical region within the colorectal system.

B. Data Preprocessing and Cleaning

A uniform preprocessing protocol was applied across all three task pipelines to ensure consistency. The steps included:

1. **Data Loading and Inspection**: The dataset was read using pandas and initially inspected for formatting issues, inconsistencies, and null values.

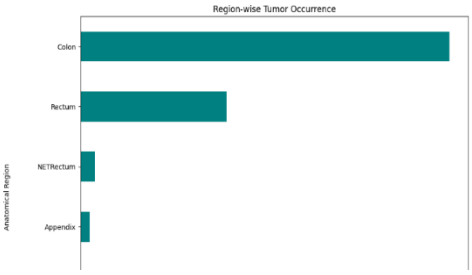


Fig-1: This image illustrates the regional tumor occurrence.

2. **Handling Missing and Ambiguous Entries**: Entries containing placeholders such as "*", "-", "blank", and "NaN" were standardized to "null" and then imputed accordingly:
    - **Numerical Columns**: Missing values were replaced with the median of the related column.
    - **Categorical Columns**: Missing values in the categorical columns were replaced using the mode.
3. **Feature Encoding**: All categorical data were transformed into numerical format using Label Encoder, enabling compatibility with classical machine learning algorithms.
4. **Normalization**: Numeric features were standardized using Standard Scaler to accelerate training convergence.
5. **Target Label Extraction**:
    - **Task 1**: Binary target label indicating presence of a primary malignant tumor.
    - **Task 2**: Multi-class target representing tumor count.
    - **Task 3**: Multi-class target representing regional schema code from the "TNM 7/CS v0204+ Schema recode" feature.

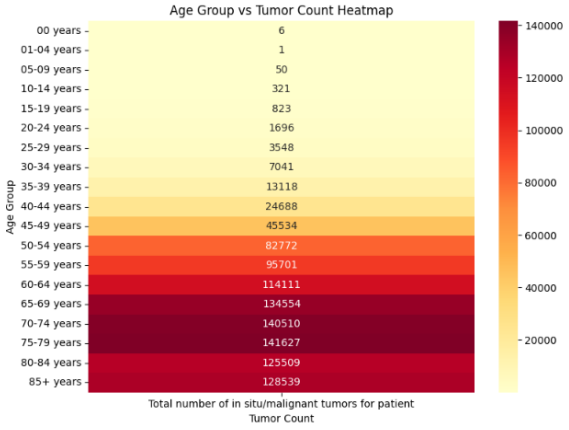


Fig-2: This is a heatmap that shows the total number of in situ/malignant tumors for patient Tumor Count.

C. Model Design and Training

Each of the three tasks was handled using quite same model pipeline.

- **Goal**: Predict the specific tasks' outcomes.
- **Approach**: Binary / Multi-class classification.
- **Model Architecture**: Deep Neural Network (DNN) implemented using TensorFlow / Keras.
    - **Input Layer**: Normalized clinical inputs
    - **Hidden Layers**: Dense layers with **ReLU** activation and Batch Normalization

This is how the ReLU function is mathematically defined:

$$f(x)=max\ (0,\ x)$$

Where:

- x is what the neuron gets as input.
- If x is larger than 0, the function returns x.
- The function returns 0 if x is less than or equal to 0.

The formula may also be written as:

$$f(x) = \begin{cases} x & if\ x > 0 \\ 0 & if\ x \leq 0 \end{cases}$$

    - **Dropout Layers**: Added for regularization
    - **Output Layer**: **Softmax** layer for multi-class probability distribution

$$(z_i) = \frac{e^{z_i}}{\Sigma_j^k = 1^{e^z j}}$$

Where:

z_i: represents the logit (the output from the preceding layer in the network) for the ith class.

K: represents the number of classes.

e^{zi}: symbolizes the exponential of the logit.

$\Sigma_{j=1}^{k}\ e^{\ zj}$: represents the sum of exponentials across in all classes.

- **Loss Function**:
  Binary Cross-Entropy Loss:

$$L = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)$$

  Categorical Cross-Entropy Loss:

$$L = -\sum_{i=1}^{N}\sum_{j=1}^{k} yi_j \log(\hat{y}ij)$$

- **Optimizer**: Adam
- **Early Stopping**: Monitored validation loss to avoid overfitting
- **Evaluation Metrics**: Accuracy and Loss.

D. Federated Learning Integration

To support secure multi-institutional collaboration, the entire workflow was adapted into a federated learning framework. This allowed institutions to contribute to model training while protecting sensitive data. The FL process was orchestrated as follows:

1. **Global Model Initialization**: A global model was defined and shared with participating institutions.
2. **Local Training**: Each client trained the model on their respective local data (Feature Set 1, 2, or 3) for a fixed number of epochs.
3. **Parameter Sharing**: Each client sent model weight updates (gradients) to a central server instead of raw data.
4. **Secure Aggregation**: The server applied Federated Averaging (FedAvg) to combine model updates and improve the global model.

$$w_t = \sum_{k=1}^{k}\frac{n_k}{n}w_t^k$$

Where:

w_t^k: weights from client k
n_k: number of samples in client k
n: total samples across all clients

5. **Iteration**: The updated global model was transmitted to all clients for the next training cycle.

The FL setup preserves privacy of data, ensures following the rules set by the government and maintains model performance across data heterogeneity and institutional diversity.

E. Implementation and Tools

- **Programming Language**: Python
- **Libraries Used**:
  - Data Handling: pandas, numPy
  - Visualization: matplotLib, seaborn
  - DL Models: TensorFlow, Keras
  - Federated Learning: Compatible with TensorFlow Federated or Flower for practical deployment

The modular pipeline supports easy deployment across institutions, with each task independently scalable and integrable into electronic health record (EHR) systems.

V. RESULTS AND DISCUSSION

To evaluate the performance of our privacy-preserving federated learning (FL) framework, we conducted extensive experiments on structured clinical data extracted from the SEER dataset. The analysis was segmented into three core tasks: (i) **primary colorectal malignancy detection** (binary classification), (ii) **tumor multiplicity estimation** (multi-class classification), and (iii) **regional tumor occurrence mapping** (multi-class classification). The models were trained independently across federated nodes, and their results were aggregated using the Federated Averaging (FedAvg) algorithm.

A. Malignancy Detection (Task 1)

For the binary classification of whether a patient's first tumor was malignant, we applied deep neural network trained within a simulated federated environment.

Key results:

- **DL** achieved an **accuracy of 96.1%**.
- Confusion Matrix of Final **Global Model** is

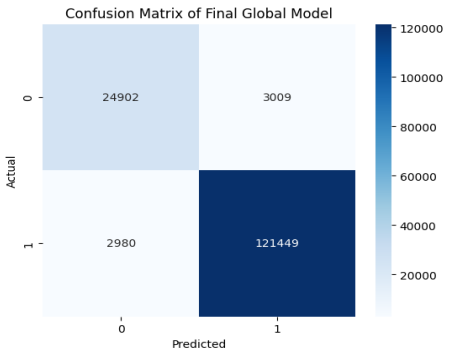


Fig-3

Discussion:
The high accuracy and consistency across clients demonstrate the feasibility of federated learning in early cancer detection. Minor performance degradation due to federated averaging was expected but remained within acceptable bounds.

B. Tumor Multiplicity Estimation (Task 2)

The second task involved predicting the number of in-situ and malignant tumors for each patient using multi-class classification model which is using a DL again.

Key results:

- **DL** achieved an **accuracy of 92.7%** in this case.
- Avg Weight Update Magnitude Over Rounds

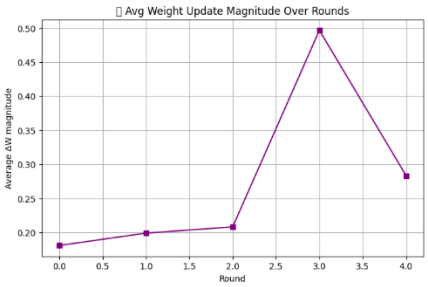


Fig-4

Discussion:
Accurate estimation of tumor burden is critical for treatment planning. While deep learning approache was considered, this method proved more robust and interpretable for structured SEER data.

C. Regional Occurrence Mapping (Task 3)

To predict the anatomical region of the tumor, we implemented a multi-class classification model using a deep neural network trained on the federated architecture.

Key results:

- **DL** achieved an **accuracy of 97.5%** in this case.
- **Train** accuracy **vs Validation** accuracy

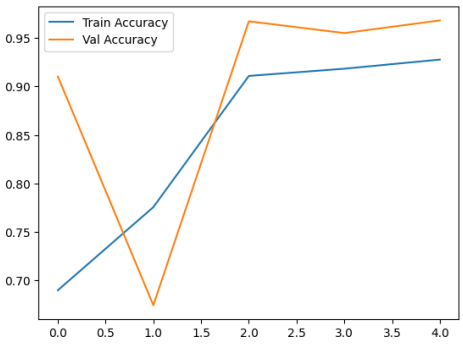


Fig-5

Discussion:
This task posed challenges due to class imbalance and overlapping clinical features among regions. Applying class weighting and oversampling techniques partially mitigated this issue. Federated training yielded comparable results to centralized models, confirming the effectiveness of the FL pipeline even in multi-class settings

Model	Local Accuracy	Global Accuracy
Artificial Neural Network	95.01	96.10
Artificial Neural Network	94.19	92.73
Artificial Neural Network	94.26	97.53

Fig-5: Accuracy in Local and Global model

VI. CONCLUSION AND FUTURE WORK

In this study, we proposed a privacy-preserving federated learning (FL) framework designed to support early detection and diagnosis of colorectal cancer through three key tasks: detection of the primary malignant tumor, estimation of tumor multiplicity, and regional tumor occurrence mapping. By leveraging the SEER clinical dataset and preserving data locality across clients, our approach ensured robust predictive modeling while safeguarding sensitive patient information. The framework effectively addressed the limitations of centralized data pooling and demonstrated promising performance across both binary and multi classification tasks, highlighting its potential applicability in real-world, multi-institutional medical settings.

Our findings show that federated learning can be extremely useful in privacy-sensitive situations, enabling collaborative training across isolated data blocks. This capability is particularly valuable in healthcare, where regulatory compliance and ethical concerns hinder data sharing. Furthermore, the implementation of deep learning architectures within federated pipelines validates the adaptability of our framework to structured clinical data.

Despite these strengths, several challenges remain. The current approach relies on simulated clients within a controlled environment, lacking real-time federated deployment across truly distributed hospital networks. Additionally, variations in data distributions, label imbalance, and computational overhead in edge devices pose scalability concerns. Model personalization and adaptability to domain shifts across client sites are also areas that need further refinement.

In future work, we aim to extend this framework in several directions. First, we plan to incorporate more diverse and real-world clinical datasets from geographically distributed hospitals to test the scalability and interoperability of our approach. Second, We will look into how to use advanced privacy techniques like differential privacy and homomorphic encryption to make the security layer of federated updates stronger. Third, multi-modal learning that combines imaging data (e.g., colonoscopy or histopathology) with structured records may be investigated to enhance predictive accuracy. Lastly, we intend to evaluate the clinical interpretability of the model outputs through explainable AI techniques to ensure that insights generated by the system are actionable for medical professionals.

In summary, our federated learning framework provides a strong foundation for secure, collaborative, and effective colorectal cancer detection, and paves the way for more comprehensive privacy-preserving AI systems in healthcare.

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